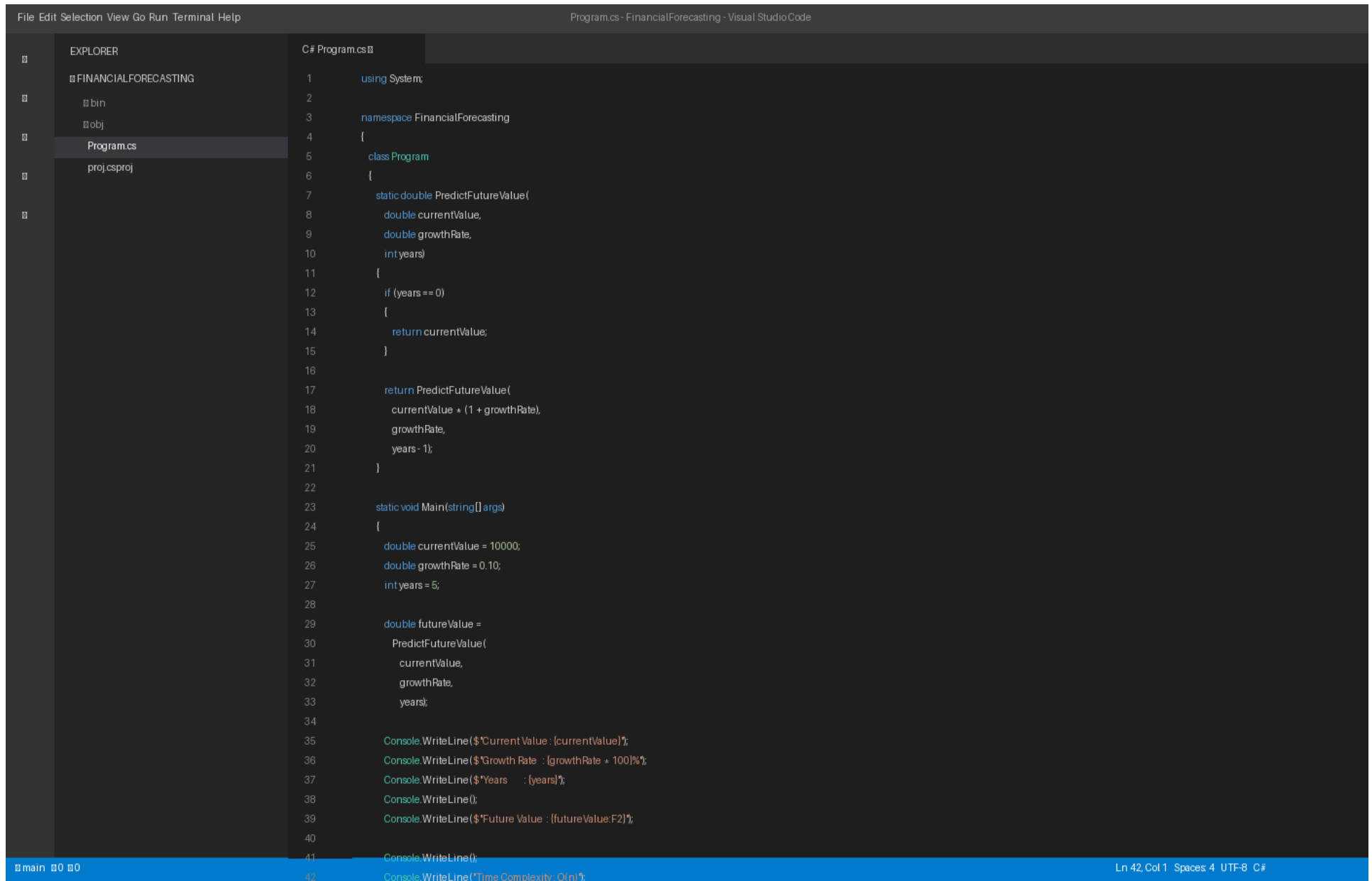


Exercise - Financial Forecasting (Program.cs)



The screenshot displays the Visual Studio Code interface for a project named "FinancialForecasting". The Explorer sidebar on the left shows the project structure, including a "bin" directory, an "obj" directory, and a "Program.cs" file. The main editor area shows the content of "Program.cs", which is a C# program for financial forecasting. The code includes a namespace "FinancialForecasting", a class "Program", and a static method "PredictFutureValue". The "Main" method initializes variables for current value, growth rate, and years, and then calls "PredictFutureValue" to calculate the future value. The output is displayed in the console at the bottom.

```
1 using System;
2
3 namespace FinancialForecasting
4 {
5     class Program
6     {
7         static double PredictFutureValue(
8             double currentValue,
9             double growthRate,
10            int years)
11         {
12             if (years == 0)
13             {
14                 return currentValue;
15             }
16
17             return PredictFutureValue(
18                 currentValue * (1 + growthRate),
19                 growthRate,
20                 years - 1);
21         }
22
23         static void Main(string[] args)
24         {
25             double currentValue = 10000;
26             double growthRate = 0.10;
27             int years = 5;
28
29             double futureValue =
30                 PredictFutureValue(
31                     currentValue,
32                     growthRate,
33                     years);
34
35             Console.WriteLine($"Current Value : {currentValue}");
36             Console.WriteLine($"Growth Rate : {growthRate * 100}%");
37             Console.WriteLine($"Years : {years}");
38             Console.WriteLine();
39             Console.WriteLine($"Future Value : {futureValue:F2}");
40
41             Console.WriteLine();
42             Console.WriteLine($"Time Complexity : O(n)");
```

Ln 42, Col 1 Spaces: 4 UTF-8 C#

Program Output (Terminal)

File Edit Selection View Go Run Terminal Help

Program.cs - FinancialForecasting - Visual Studio Code

EXPLORER

FINANCIALFORECASTING

bin

obj

Program.cs

proj.csproj

C# Program.cs

```
1 using System;
2
3 namespace FinancialForecasting
4 {
5     class Program
6     {
7         static double PredictFutureValue(
8             double currentValue,
9             double growthRate,
10            int years)
11         {
12             if (years == 0)
13             {
14                 return currentValue;
15             }
16
17             return PredictFutureValue(
18                 currentValue * (1 + growthRate),
19                 growthRate,
20                 years - 1);
21         }
22
23         static void Main(string[] args)
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PSC:\Users\Student\FinancialForecasting> dotnet run
Current Value : 10000
Growth Rate : 10%
Years : 5

Future Value : 16105.10

Time Complexity: O(n)
Optimization: Use Math.Pow() to avoid recursive calls

PSC:\Users\Student\FinancialForecasting>
```

main 00 00

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